



Assessing the compressive and splitting tensile strength of self-compacting recycled coarse aggregate concrete using machine learning and statistical techniques

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ABSTRACT

The construction industry is adopting high-performance materials due to technological and environmental advances. Researchers worldwide are studying the use of recycled coarse aggregates (RCA) as a partial alternative to natural coarse aggregates in concrete due to their sustainability and environmental benefits. This study compares the predictive abilities of three different machine learning techniques in evaluating the mechanical properties of 28-day-old self-compacting concrete (SCC) incorporating RCA to better understand how design parameters affect the mechanical properties of SCC containing RCA. The study used a range of statistical methodologies and machine learning algorithms, such as ANN, SVM, and M5P trees, to examine the relationship between design elements and accurately forecast the mechanical characteristics of concrete. The ANN model exhibited notable superiority in effectively forecasting compressive strength (CS) and splitting tensile strength (STS) compared to other models, with uncertainty bands of 15.038%–21.154% for CS and 15.701%–19.008% for STS. Moreover, all the uncertainties were under the threshold of 35%. Notably, the water-cement ratio emerged as the most crucial parameter in accurately predicting the mechanical properties of SCC. Finally, the parametric evaluation conducted revealed that CS and STS are inversely proportional to water-cement ratio and aggregate-cement ratio, whereas, are directly proportional to the water-binder ratio, and water-solids percentage.

1. Introduction

The introduction of waste legislation, encompassing guidelines and directives, in numerous countries worldwide has led to a substantial transition towards environmental-friendly sustainability practices and the effective management of construction and demolition waste (CDW). This shift has evolved from a mere recommendation to a legal obligation and a necessary course of action. Consequently, various sectors within the construction industry including building, infrastructure, and industrial, are formulating and implementing strategies to minimize waste generation and enhance the management of CDW. These efforts are undertaken with the overarching goals of safeguarding the environment and fostering economic sustainability [1,2]. The construction industry

plays a significant role in both material consumption and waste generation [1–3]. Notably, concrete accounts for a substantial 70% of CDW [4]. In light of the increasing frequency of building demolitions, it has become imperative to explore methods for reusing this waste to conserve finite resources. In light of this challenge, the concept of sustainable and environmentally friendly concrete, characterized by its utilization of recycled coarse aggregate (RCA) has gained prominence as a construction material [5,6]. Sustainable concrete, often referred to as "green" concrete, strives to achieve several key objectives. It promotes recycling, reduces energy and natural resource usage, avoids environmental harm, and actively promotes ecological sustainability and development.

Consequently, As a result, the full potential of RCA within the concrete industry faces several intricate technological challenges. It is

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imperative to acknowledge that each particle of RCA remains a composite entity, comprised of recycled concrete aggregate with its inherent natural coarse aggregate (NCA) and an adhered mortar layer. RCA essentially emerges from amalgamating aged concrete fragments with a fresh matrix. To gain a comprehensive understanding of RCA and establish industry standards, it becomes essential to distinctly identify and discern the components of these composite particles, which has grabbed the attention of many researchers in recent years [7] (Fig. 1a). This differentiation is crucial for making informed predictions regarding the potential impact of RCA on concrete. The final product's quality is mostly determined by the RCA's quality, the details of the mix design, the mixing technique used, the percentages of replacement (refer to Fig. 1b), and the RCA's condition. It is worth noting that much of the research pertaining to RCA's utilization tends to emphasize its mechanical properties. Although it is theoretically possible to employ RCA as an NCA, it is widely accepted that doing so frequently results in a loss in RCA strength [8]. To remedy this scenario, several measures may be taken such as increasing the cement content, modifying the water-cement ratio (w/c), and including supplementary cementitious materials (SCMs) into the concrete mix design [9]. Over the past several decades, concrete production has seen significant growth, highlighting its versatility as a building material. One of the critical properties of concrete is its compaction process, which imparts superior durability and strength to the material. Self-consolidating concrete (SCC) is a type of concrete that has improved flow ability and the ability to fill complex reinforcement and formwork. This alternative to traditional concrete can be cast and compacted using weight alone, eliminating the need for external vibrations. SCC has several enhanced characteristics that make it a valuable alternative to traditional concrete in specific construction applications. Appropriate filling of structural performance is facilitated by SCC [10]. SCC was proposed for the first time in Japan in 1988 [11] and since then has been recognized in construction practices [12].

Within the construction field, researchers and engineers have undertaken extensive investigations into methods to enhance the strength and durability of SCC [13]. These endeavours have led to the proposal of diverse approaches for formulating SCC mixes, employing different

mixing techniques to attain high-performance self-compacting concrete (HP-SCC) tailored to meet specific requirements [14,15]. HP-SCC is characterized by its notably high compressive strength (CS) [14,15]. Typically, achieving such properties involves the utilization of elevated cement content and a reduced w/c, which can be further optimized through the incorporation of inert or active SCMs into the mix [16]. In light of the growing demand for sustainable construction materials exhibiting robust mechanical attributes to ensure prolonged service life, prior research efforts have explored various design parameters pertinent to the mechanical properties. Therefore, a thorough assessment of the existing level of knowledge in this field is required to increase our understanding of concrete design requirements. This revised study should give a thorough review of the advancements achieved in the sector and help develop design standards. Furthermore, it's crucial to comprehend the connections between the design criteria and the qualities of the final hardened concrete. When considering the design parameters for concrete mixes aimed at achieving superior quality, it becomes imperative to gather comprehensive experimental data. Data from the vast body of literature on SCC should be analyzed rigorously to determine the relationships between all possible pairings of concrete design factors [17]. Such an approach affords a profound and interdisciplinary understanding of the correlations between each pair of variables. It provides scientifically rigorous clarity regarding experimental outcomes, enabling the potential estimation of other variables based on established relationships. Using SCC concrete mixtures makes the CS and STS of concrete more problematic and complex to be estimate; consequently, more advanced computing approaches are required to create predictive models [18].

Numerous parameters contribute to the determination of SCC properties, with design parameters emerging as particularly influential. The majority of concrete's characteristics hinge on the precise proportions of ingredients incorporated into the mixture. Even slight deviations in these proportions can lead to significant disparities in both fresh and hardened concrete attributes, as substantiated by existing literature. Researchers frequently manipulate key ratios, such as w/c and water-to-binder (w/b) ratios [15]. Beyond these fundamental ratios, various other proportional relationships have been identified as direct influencers of concrete properties, as documented in the literature [8,15,19]. Collectively, these proportional values, percentage compositions, or ingredient masses used in the concrete mixture are collectively termed design parameters. Remarkably, irrespective of mixing conditions or environmental factors, these design parameters exert considerable control over concrete properties. Fig. 2 underscores the pivotal role of solid materials, particularly in the modification of recycled coarse aggregate (RCA) concrete, where the ratios of coarse aggregate to fine aggregate (FA/CA), total aggregate to cement (a/c), and water to solid (w/s) ratios assume significant importance.

In the field of materials science, various soft computing approaches or machine learning (ML) have been adopted for predicting the CS and STS of concrete, such as artificial neural networks (ANNs), standard neural networks (SNNs), and Bayesian neural networks (BNNs) [17,18]. The literature contains numerous in-depth reports on using ANNs to model concrete materials [17,18]. In addition to ANN models, other computational techniques such as fuzzy logic (FL), adaptive neuro-fuzzy inference systems (ANFIS), fuzzy functions with least squared estimation (FF-LSE), genetic algorithms (GA), linear regression (LR), bagging regression trees (BRT), non-linear regression (NLR), multiple regression (MR), multiple additive regression trees (MART), hybridized particle swarm optimization (PSO), support vector machines (SVM), and M5P trees have also been used in recent years to model the mechanical properties of concrete materials [18,20,21]. Literature indicates that most research employed either conventional or hybrid ANNs to predict the CS and STS of concrete. Furthermore, the offered models' practical applications were not given enough attention. Conventional ANNs have been used in much earlier research for analysis; however, hybrid soft computing models have the ability to overcome drawbacks such as over

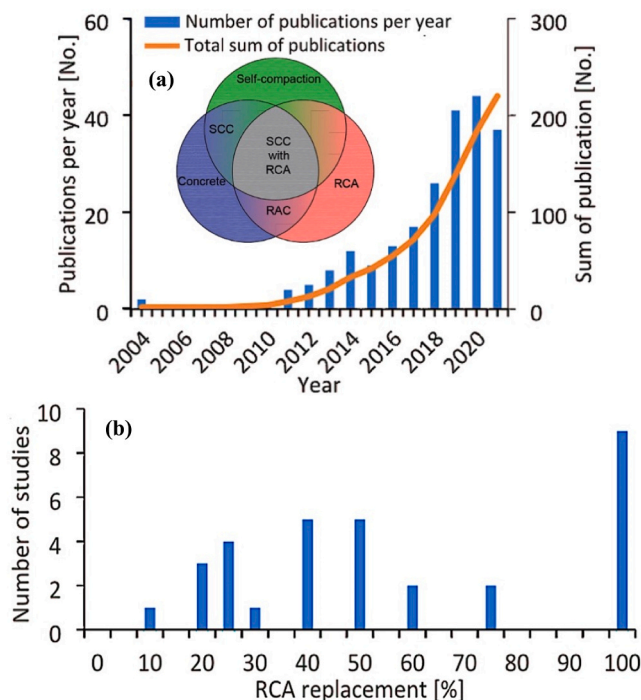


Fig. 1. SCC: (a) publications' number in Scopus database and schematic motivation and classification (b) RCA replacement %. (Copyright permission was taken from [7]).

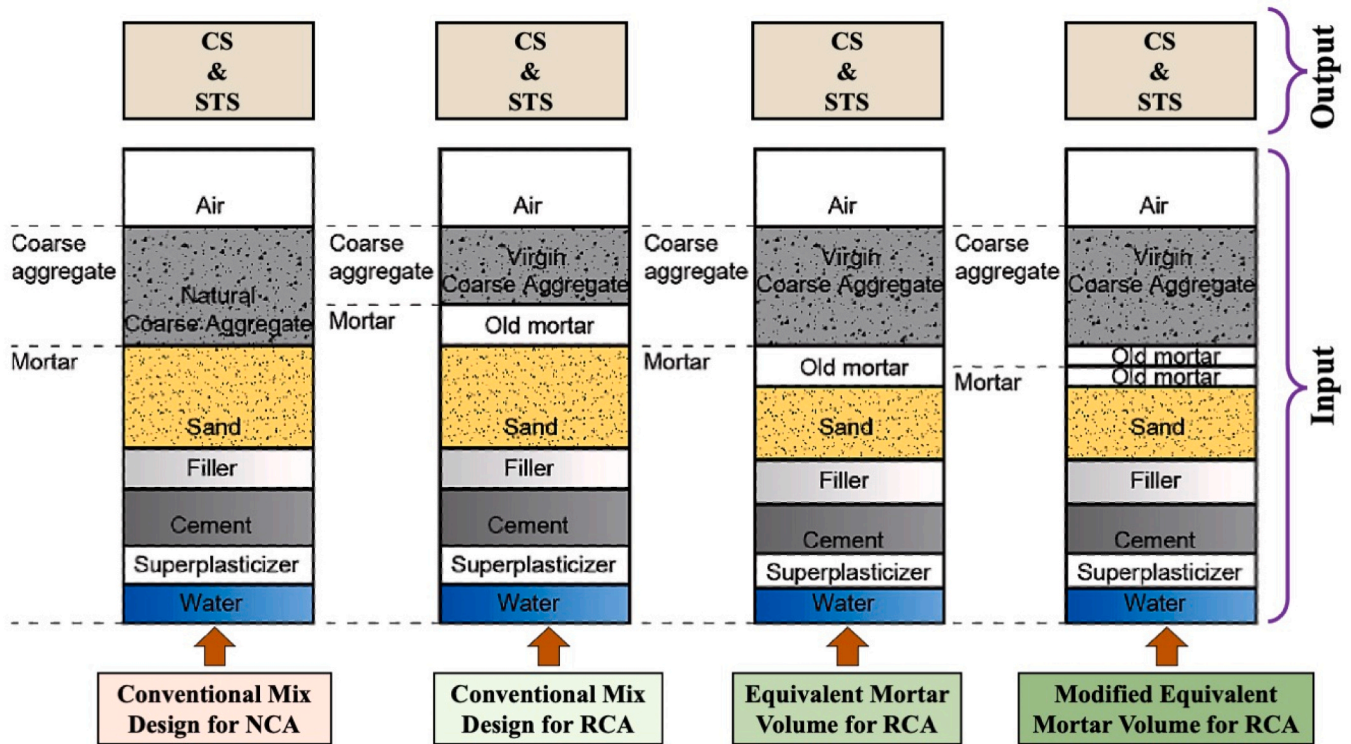


Fig. 2. Schematic representation of SCC compositions: highlighting their design variations using diverse mix design methodologies in contrast to the conventional approach.

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fitting and local minimum difficulties in forecasting the desired output. However, considerable empirical modeling to address these concerns has mostly been done on big datasets, while traditional ANNs have proven a more impressive capacity to reliably anticipate outputs on small datasets, as seen in Table 1. Given that ANNs accurately predict the CS and STS of materials, it is important to examine how design parameters influence desired strengths and which elements have the most impact on the CS and STS of SCC. One of the many contemporary soft computing strategies being researched is the use of ANNs to evaluate and forecast certain concrete characteristics, as seen in Fig. 3. The ANN makes predictions about the strengths of concrete based on data and prior experiences [22].

As shown in Table 1, there are some contradictions among studies in terms of the best prediction model; however, that is due to the used variables in the model. As a result, these factors are crucial to the prediction process. The physical, chemical, and micro properties of the materials, such as cement, admixtures, and aggregates, as published in the literature, are not included in the dataset utilized in this investigation. In order to achieve the hardened properties for the various grades of SCC, it is essential to take into account the addition of admixtures (such as viscosity-modifying agents and superplasticizer) and supplemental cementitious materials (such as ground granulated blast furnace slag, silica fume, and fly ash). Despite the very modest amount of admixture utilized in the mixture, it was nonetheless taken into account while creating the dataset. The ratios produced from the basic components (also known as non-dimensional components) are supplied for the sole purpose of modeling. Thus, this study aimed to:

1. Add new non-dimensional elements (design parameters) to the literature.
2. Investigate the controversies and contradictions surrounding the mix design parameters of SCC using pairwise comparison analysis with “variance inflation factor (VIF)” and “Pearson’s correlation

coefficient (PCC)”. The focus was on identifying pairs of input variables that show strong positive or negative correlations within the mix designs.

3. Identify the possibility of multicollinearity taking place between the independent variables, which would affect the prediction models.
4. Predict the mechanical characteristics of the material (CS and STS) using ML techniques (ANN, SVM, and MSP).
5. Assess the implemented ML-based techniques throughout indicators such as the correlation coefficient (CC), mean absolute error (MAE), root mean square error (RMSE), comprehensive measure (COM), and scatter index (IS).
6. Assess the influence that each one of the design parameters has on the targeted strengths’ predictability and then rank them via sensitivity analysis. Moreover, conduct a parametric analysis based on the results of the sensitivity analysis for better understanding (both analyses were conducted using the best predictive model).

The results of these analyses were used to provide insights and understanding for future engineers and scholars on the appropriate design parameters for SCC with RCA. The methodology for reviewing, extracting, and analyzing data is presented in Section 2. The methods used in this study are described in Section 3. The primary findings of ML application and practical insights gained from the assessment are discussed in Section 4. Limitations and future orientation are illustrated in Section 5. Finally, the paper is concluded in Section 6.

2. Reviewing, materials, and methodology

Fig. 4 presents the methodology used for this investigation. The approach to data collection involved selectively gathering and examining publications specifically related to SCC while disregarding other articles. Priority was given to SCC articles that incorporated RCA, and the remaining articles were excluded to ensure a comprehensive

Table 1

Previous works conducted ML for the prediction of strengths (CS and STS) based on dimensional and non-dimensional design parameters.

Author	Input	Models	Best model
[21]	C, W, FAgg, CA, Admixture (kg/m ³) FA, and w/b	ANNs, GA	ANN
[22]	C, BFS, W, FA, SP, CA, FAgg, and T	SNN, BNN, and LR	BNN
[23]	C, SF, W, SP, CA, FAgg, and T	LR, NLR, ANNs, ANFIS	ANNs
[24]	C, BFS, W, FA, SP, CA, FAgg, T and CS	PSO and ANNs.	ANNs
[25]	C, BFS, FA, SP, CA, FAgg, T, and CS	ANN, SVM, MR, and Tree-based (MART and BRT)	Tree-based
[26]	C, W, CA, FAgg, SF, SP, FA, GGBS	30 different architecture networks were generated	8-10-6-1 architecture ANNs
[27]	C, BFS, FA, SP, CA, FAgg, T, and CS	LR, ANNs, and SVR	SVR and LR
[28]	C, BFS, W, FA, SP, CA, FAgg, and T	SVM, ANNs, and EFSIMT	EFSIMT and ANNs
[29]	w/b, W, FAgg/CA, AC, SP	kNNA and GRNN	kNNA
[30]	Fibre (kg/m ³), C, W, SF, FA, SP, CA, FAgg, T, and CS	FL and ANNs	FL
[31]	C, SF, W, FAgg, CA, Filler (kg/m ³),	SVM and ANNs	SVM
[32]	C, SF, W, FAgg, CA, SP, WPCF	NLR, ANNs, and ANFIS	ANNs

Note “C: Cement (kg/m³), w/b: Water–binder ratio (%),FA: Fly ash (kg/m³), BFS: Blast furnace slag (kg/m³), W: Water (kg/m³), SP: Super plasticizer (kg/m³), CA: Coarse aggregate (kg/m³), FAgg: Fine aggregate (kg/m³), SF: Silica fume (kg/m³), FAgg/CA: Ratio of the weight of fine aggregate to the weight of all aggregate (%), AC: The content of air-entraining agent, EFSIMT: Evolutionary fuzzy support vector machine, GGBS: Granulated graded blast furnace slag (kg/m³), RHA: rice husk ash (kg/m³), PC30: Portland cement 30 (kg/m³), PC40: Portland cement 40 (kg/m³), WPF: Waste propylene fiber (kg/m³), T: Age of concrete (day), and CS: Compressive strength (MPa), kNNA: k nearest neighbor algorithm, GRNN: generalized regression neural network”.

investigation into the development mechanism of SCC. To enhance the essential characteristics of SCC mixtures, the use of additives was considered, not only for cost minimization but also to optimize performance. Articles that included the utilization of admixtures were accorded greater value and received more attention. The design parameters for evaluating CS and STS were derived from various mix designs documented in the literature as non-dimensional elements (design parameters). Furthermore, the evaluation used five techniques, including two statistical techniques such as PCC and VIF to analyze the correlation between independent variables and three ML techniques (ANN, SVM, and M5P) to assess the datasets.

2.1. Extraction of datasets

Mix design is a critical process involving the precise determination of ingredient quantities (Kg/m³), to achieve specific properties in hardened SCC. In a comprehensive analysis, a total of 210 published mix designs were scrutinized, focusing on several key design parameters. These parameters encompassed the $\frac{w}{c}, \frac{w}{b}, \frac{a}{c}, \frac{FA}{CA}, \frac{w}{s}$, and RCA%, all expressed in non-dimensional units as percentages, as well as the superplasticizer (SP) quantity, expressed in dimensional units as kilograms. The selection of these design parameters was guided by their prevalence in mixed-design approaches and the availability of data within the existing literature, with an emphasis on comprehensive studies that included a wide range of components. Studies with limited substance variation were excluded from the analysis. To ensure a meaningful analysis, the dataset was further categorized based on the CS grade of SCC. Moreover,

the data was stratified based on both CS and STS measurements taken after 28 days of curing, without regard to variations in specimen size, shape, or testing standards employed. A comprehensive overview of the dataset’s range is presented in Table S1, available in the supplementary information.

2.2. Design parameters in literature and statistical evaluation

Table 2 presents a compilation of 23 studies from Q1 journals indexed in Scopus only for more reliability, focusing on CS and STS of SCC incorporating RCA; CS and STS ranged from 13 to 80 MPa and 1 to 7 MPa, respectively, at 28 days. This paper provides details about 210 different mixture proportions.

Table 3 presents statistical summaries for seven inputs ($\frac{w}{c}, \frac{w}{b}, \frac{a}{c}, \frac{FA}{CA}, \frac{w}{s}$, SP(kg), and RCA%) plus two outputs (CS and STS). The data analysis in Table 3 shows that the compressive strength varied between 13.64 and 81.40 MPa. Based on the collected data, the average compressive strength was found to be 43.925 MPa, with a standard error (SE) of 0.845 MPa and a standard deviation (SD) of 12.244 MPa. The values ranged from 0.96 to 7.20 MPa for the splitting tensile strength. Based on the data obtained, the mean splitting tensile strength was calculated as 3.409 MPa, with a SE of 0.064 MPa and a SD of 0.922 MPa.

The results gathered demonstrated that the input variables significantly influence the evaluated strength parameters of the concrete mixture. Figs. 5 and 6 show a parallel correlation between specified input factors and concrete strength performance (CS and STS) to better understand their involvement. The highest CS of 81.40 MPa was obtained for a combination of input variables as RCA% = 0, $\frac{w}{c}$ = 0.3, $\frac{w}{b}$ = 0.2, $\frac{a}{c}$ = 4.34, $\frac{FA}{CA}$ = 2.04, SP = 7, $\frac{w}{s}$ = 5.11. While the lowest CS of 13.64 MPa was obtained for a combination of input variables as RCA% = 100, $\frac{w}{c}$ = 0.35, $\frac{w}{b}$ = 0.09, $\frac{a}{c}$ = 12.44, $\frac{FA}{CA}$ = 6.04, SP = 7.4, $\frac{w}{s}$ = 2.13. The highest STS of 7.20 MPa was obtained for a combination of input variables as RCA% = 0, $\frac{w}{c}$ = 0.42, $\frac{w}{b}$ = 0.25, $\frac{a}{c}$ = 3.75, $\frac{FA}{CA}$ = 2.10, SP = 4, $\frac{w}{s}$ = 7.8. While the lowest STS of 0.96 MPa was obtained for a combination of input variables as RCA% = 100, $\frac{w}{c}$ = 0.35, $\frac{w}{b}$ = 0.09, $\frac{a}{c}$ = 12.44, $\frac{FA}{CA}$ = 6.04, SP = 7.4, $\frac{w}{s}$ = 2.13..

The performance of the specified concrete changes when one variable changes. Consequently, determining the ideal combination of input variables to produce a concrete mixture with the best performance is challenging. Notably, the strength increases when the w/c or w/b ratio decreases. All high-strength mixtures examined in the studies exhibited an a/c ratio ranging from 3.11 to 24.17. The FA/CA ratio ranged from 1.13 to 9.87, while the SP dosage varied from 1.16 to 10.8 kg. A comparative analysis of the findings depicted in both Fig. 5 and Fig. 6 reveals a notable trend: as the ratios of a/c and FA/CA decrease, there is a corresponding increase in the concrete’s strength. Interestingly, as the percentage of RCA increases, so do the utilization percentages of the w/s ratio. Conversely, there is a decrease in the frequency of SP usage when RCA replacements are absent. To compensate for the reduced SP usage in these cases, there has been an elevation in the w/c ratio. Consequently, the w/s ratio within the system is higher than observed in earlier ranges, owing to a significant reduction in added water. It’s noteworthy that in some studies, the w/c and w/b ratios exceeded conventional values, spanning a range from 0.09 to 0.5.

3. Evaluation techniques and assessment criteria

In this work, five computational approaches have been implemented to assess datasets: PCC and VIF (statistical), ANNs, SVM, and M5P (ML-based). Moreover, datasets underwent a single-factor analysis of variance (ANOVA) to discern whether there exists a statistically significant difference or insignificance between the predicted and observed values produced by various soft computing approaches.

First, the Pearson Correlation Coefficient (PCC), often symbolized as ‘r,’ serves as a widely employed metric for gauging the linear

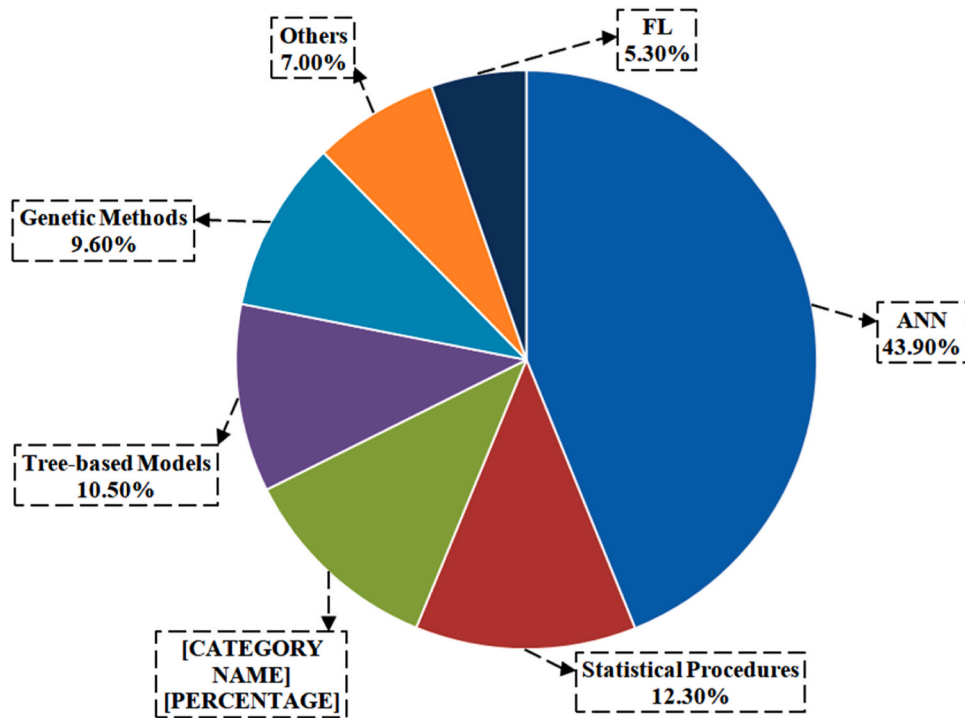


Fig. 3. Overview of proportional soft computing utilization in 65 peer-reviewed papers derived from reliable journals [33].

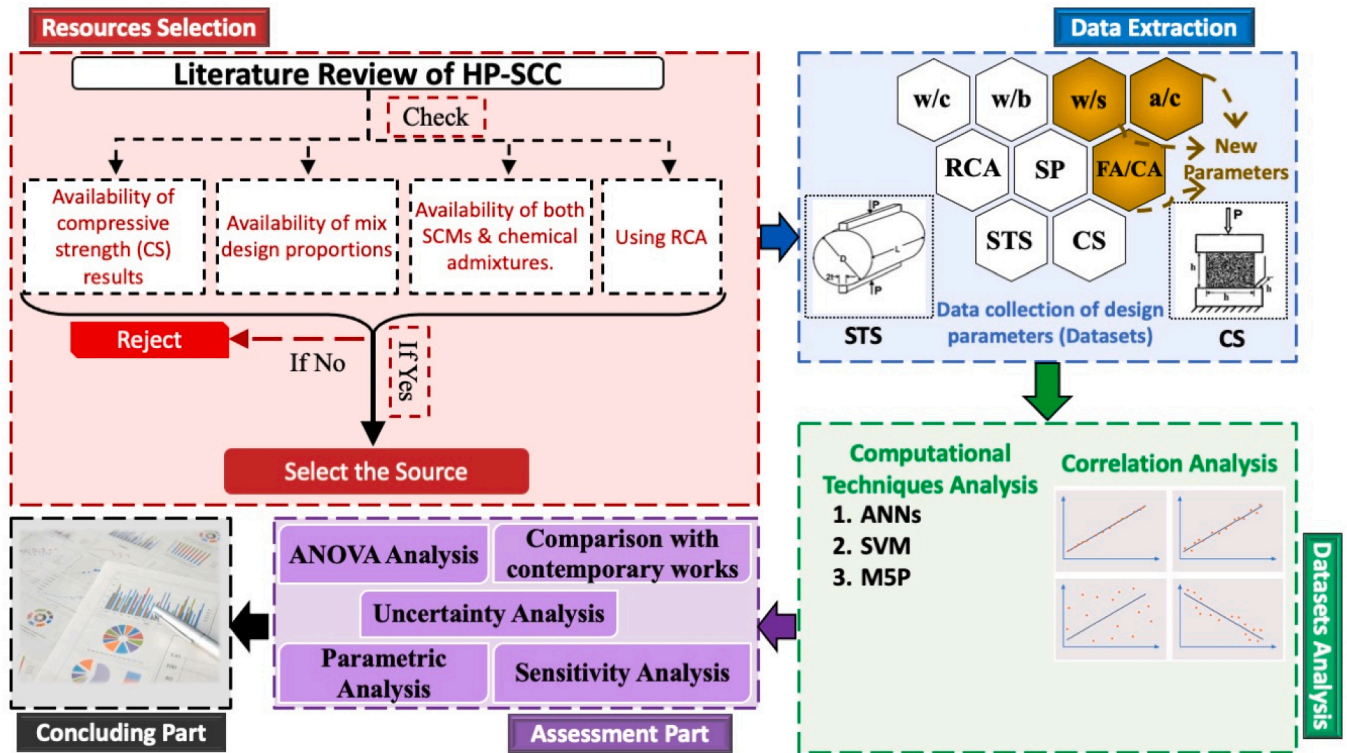


Fig. 4. Methodology of reviewing the literature.

relationship between two variables. Ranging from -1 to 1 , PCC not only quantifies the strength of this correlation but also delineates its direction [57,58]. In this study, PCC was the foundational tool for analyzing and discerning correlations among each pair of design parameters. The computation of PCC adheres to a standard formula, typically expressed as Eqs.1–3;

$$PCC = \frac{\text{covariance}(x, y)}{\sqrt{\text{var}(x)}\sqrt{\text{var}(y)}} \quad (1)$$

Table 2

Mix design parameters of SCC from the literature (number of extracted data from each study).

S. No.	Author	Number of data	S. No.	Author	Number of data
no.1	Gesoglu et al. [34]	16	no.13	Martínez-García et al. [35]	4
no.2	Sadeghi-Nik. et al.[36]	12	no.14	Duan et al. [37]	10
no.3	Revilla Cuesta et al.[38]	5	no.15	Pan et al.[39]	6
no.4	Fiol et al.[40]	12	no.16	Manzi et al. [41]	4
no.5	Señas et al. [42]	6	no.17	Nili et al.[43]	9
no.6	Aslani et al. [44]	15	no.18	Singh et al. [45]	12
no.7	Guo et al.[46]	9	no.19	Bahrami et al. [47]	4
no.8	Uygunoğlu et al.[48]	8	no.20	Silva et al. [49]	5
no.9	Sun et al.[50]	10	no.21	Grdic et al. [51]	3
no.10	Surendar et al. [52]	7	no.22	Tang et al. [53]	5
no.11	Babalola et al. [54]	14	no.23	Tuyan et al. [55]	12
no.12	Nieto et al.[56]	22			Total= 210

Where : $covariance(x,y) = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{n - 1}$ (2)

$var(x) = \frac{\sum_{i=1}^n (x_i - \bar{X})^2}{n - 1}$ (3)

Where: $\bar{X}, \bar{Y}$ = mean values, n =datasets.

Second, Multicollinearity manifests when notable interdependencies surface among two or more independent variables within the intricate framework of multiple regression. The intricate structure of this phenomenon might create bias or modify results, which presents a difficulty for researchers in their efforts to determine the best use of individual factors for predicting or understanding the dependent variable. It is important to note that examining the CC alone, as discussed in Section 3.1, provides only a limited perspective on how an individual input variable influences the output. This approach does not consider how that variable might interact with all the other variables simultaneously. In essence, it falls short in revealing the broader picture of multicollinearity-how an independent variable correlates not only with

one variable but with all other variables together. Consequently, it becomes imperative to calculate the VIF to gain a comprehensive understanding of multicollinearity through the utilization of Eq. 4 [59–61].

$VIF_i = \frac{1}{(1 - CC_i)}, i = 1, \dots, k$ (4)

Where:

"k" representing independent variables.

"CCi" stands for the "coefficient of determination", computed by fitting the regression with ith as the target while still at the same time using all other independent variables.

3.1. Assessment criteria

The effectiveness of the modeling was assessed using the previously mentioned performance metrics: CC, MAE, and RMSE [62]. These metrics are widely recognized within the statistical community for evaluating the efficacy of computational approaches. They were applied to a dataset split into 70% for training and 30% for testing, offering a comprehensive evaluation of model performance. The choice of these metrics was deliberate for two reasons:

1. Alignment with existing literature: These metrics are commonly employed in research and align with established practices in the field.
2. Suitability: They are well-suited to the nature of the data and strike a balance between simplicity and comprehensiveness in assessing model performance.

It is of paramount importance to highlight the importance of presenting a diverse set of metrics rather than relying on just one. The use of a multidimensional technique aims to effectively tackle the inherent limits associated with particular indicators, hence offering a more thorough evaluation of model performance. Within the realm of these metrics, an elevated CC, in conjunction with diminished MAE and RMSE values, signifies heightened model precision. The computations for these metrics are elaborated upon in Eqs. 5–7.

$CC = \frac{\sum_{i=1}^N (O_i - \bar{O})(P_i - \bar{P})}{\sqrt{\sum_{i=1}^N (O_i - \bar{O})^2} \sqrt{\sum_{i=1}^N (P_i - \bar{P})^2}}$ (5)

$MAE = \frac{1}{N} \sum_{i=1}^N |P_i - O_i|$ (6)

Table 3

Statistical description of the data.

	Inputs							Outputs	
	w c	w b	a c	FA CA	SP(kg)	w s (%)	RCA (%)	CS(MPa)	STS(MPa)
Mean	0.522	0.337	5.400	2.738	4.069	7.838	44.352	43.925	3.409
SE	0.017	0.006	0.220	0.094	0.143	0.139	2.693	0.845	0.064
Median	0.450	0.340	4.325	2.264	3.825	8.100	40.000	41.625	3.300
Mode	0.400	0.450	3.488	1.884	7.000	8.380	0.000	42.340	3.200
SD	0.243	0.086	3.183	1.358	2.075	2.011	39.026	12.244	0.922
Sample Variance	0.059	0.007	10.128	1.844	4.305	4.046	1523.024	149.909	0.849
Kurtosis	4.145	0.421	10.450	5.642	-0.172	0.356	-1.405	0.569	1.918
Skewness	2.148	-0.766	3.009	2.245	0.542	-0.499	0.345	0.762	0.845
Range	1.390	0.410	21.057	8.643	9.640	9.500	100	67.760	6.240
Minimum	0.250	0.090	3.110	1.229	1.160	2.130	0	13.640	0.960
Maximum	1.640	0.500	24.167	9.872	10.800	11.630	100	81.400	7.200
Count	210	210	210	210	210	210	210	210	210
Confidence level (95%)	0.033	0.012	0.433	0.185	0.282	0.274	5.309	1.666	0.125

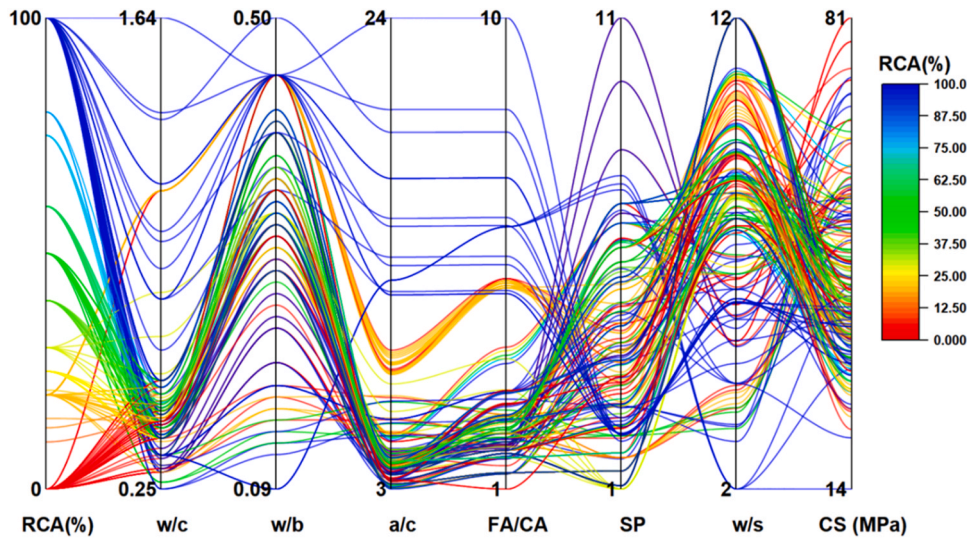


Fig. 5. Parallel correlation between the used variables and compressive strength.

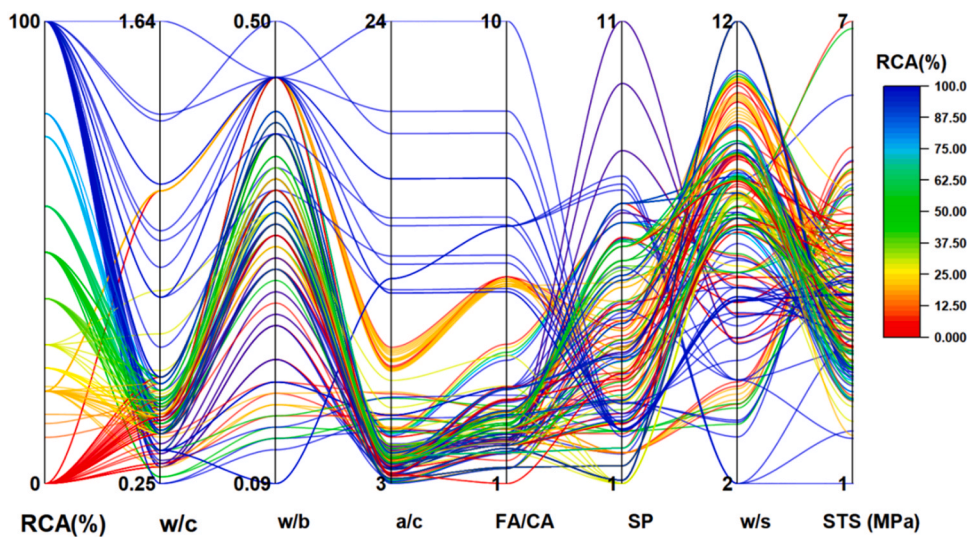


Fig. 6. Parallel correlation between the used variables and splitting tensile strength.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (P_i - O_i)^2} \quad (7)$$

More importantly, knowing that RMSE lies under the accepted range, there is a need to calculate the scatter index (SI) (Eq.8).

$$SI = \frac{RMSE}{\bar{x}} \quad (8)$$

As a diligent researcher, it is widely acknowledged that a superior-performing model is characterized by lower values of RMSE and MAE, as well as a higher value of CC. However, it's vital to acknowledge that the insights derived from these metrics may not consistently converge. Hence, there is a need for a comprehensive evaluation methodology. A holistic statistic called COM (comprehensive measure) has been established to address this difficulty and provide a uniform way for evaluating model performance. The calculation of COM incorporates the analysis of all metrics, as specified in Eq. 9, resulting in more comprehensive assessment criteria. The COM combines RMSE, MAE, and CC into a single measure. This approach helps to account for any differences that may exist among these individual metrics, resulting in a more thorough assessment of the overall performance of a model [63].

$$COM = \left(\frac{1}{3} \frac{RMSE_{Training} * MAE_{Training}}{CC_{Training}} \right) + \left(\frac{2}{3} \frac{RMSE_{Testing} * MAE_{Testing}}{CC_{Testing}} \right) \quad (9)$$

The distribution of weight for assessing performance was established with a ratio of 1/3 allocated to the training set and 2/3 to the testing set. The choice to analyze the model's performance on the testing dataset was prompted by the desire to get significant insights into its capacity to generalize. This warrants additional attention to ensure accurate and reliable results. It is imperative to underscore that a reduced value of the COM indicates superior overall model performance [63].

3.2. Machine learning (ML)

3.2.1. Artificial neural networks (ANNs)

ANNs are computational models designed to mimic the information processing and analysis capabilities of the human brain [64]. They have found extensive application in addressing numerical estimations and various challenges in the field of civil engineering [65]. Fig. 7 provides an illustration of the typical structure of an ANN, comprising an input layer, one or more hidden layers, and an output layer. The

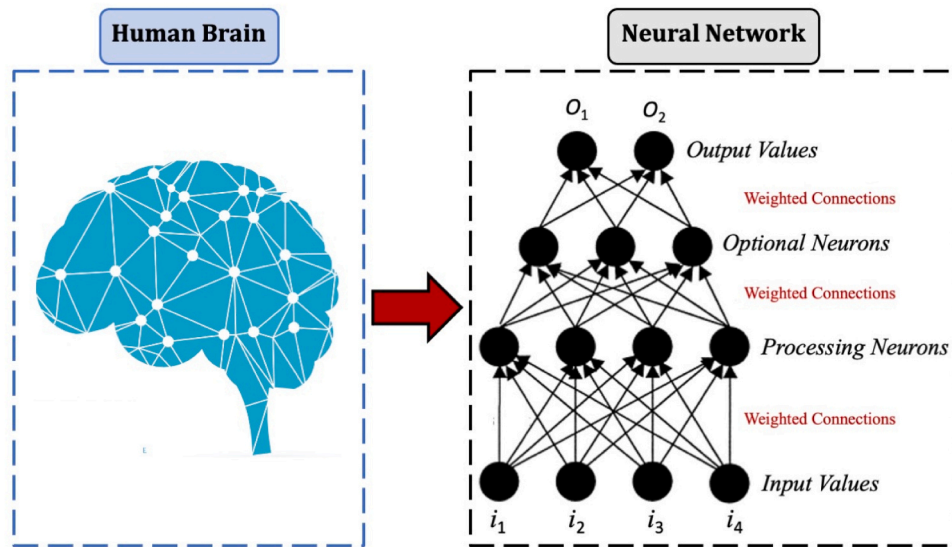


Fig. 7. Flowchart of ANNs.

interconnections between these layers involve weights, transfer functions, and biases. The determination of the number of hidden layers and neurons within the network is not arbitrary; instead, it is typically arrived at through a process of trial and error, with the MAE serving as a crucial criterion [66]. The second stage in optimizing an ANN involves identifying the ideal number of training epochs that yield the lowest values for MAE, RMSE, and CC [18].

This research investigated the use of hyperbolic tangent transfer functions in neural networks to examine how the "number of epochs" affects the reduction of RMSE and MAE. The examination focused on elucidating the fluctuations in MAE concerning varying epochs within the initial ANN models. The ANN model that was developed consisted of two hidden layers, as seen in Fig. 8. The first hidden layer consisted of five neurons, while the second hidden layer had two neurons. The model was trained with a learning rate of 0.2 and a momentum of 0.1.

3.2.2. Support vector machine (SVM)

SVM is a ML algorithm grounded in statistical principles, initially introduced by Vapnik in 2000 [67]. It has gained recognition as an innovative approach for solving classification problems and is often

recommended for such scenarios. An extension of SVM, known as Support Vector Machine for Regression (SVR), was developed by incorporating Vapnik's insensitive loss function. SVR is designed to tackle estimation challenges in non-linear regression problems. Its application has expanded beyond its original domain, finding utility in various fields, including optimal controls, regression interval analysis, and time series forecasting [68].

Nevertheless, a critical aspect of SVM or SVR lies in the selection of several parameters, termed hyper parameters [69]. The epsilon parameter, introduced by Vapnik, comprises a blend of critical components within the SVM methodology. It is important to note that SVM provides a range of kernels, including radial basis kernels (RBF) of Gaussian and Polynomials. However, it's particularly intriguing that the Pearson VII Universal Kernel (PUK) has consistently exhibited superior performance [69]. Eq. 10 serves as an illustrative representation of the core shape of the PUK, primarily geared towards its application in curve-fitting scenarios [69]:

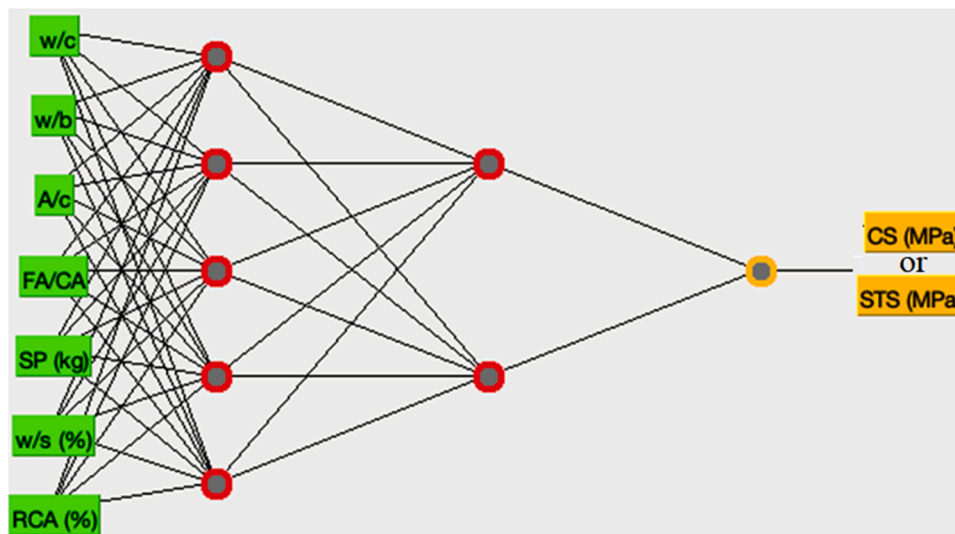


Fig. 8. Structures of the best ANN model.

$$f(x) = \frac{H}{\left[1 + \left(\frac{2(x-x_0)\sqrt{2^{\frac{1}{b}}-1}}{\sigma}\right)^2\right]^w} \quad (10)$$

Where:

H is the peak height at the middle x_0 ,
 x is the independent variable.
 ω , and σ are the parameters.

The primary motive behind employing the PUK lies in the capacity to manipulate the parameters w and σ . In this context, Eq.11 delineates the input space kernel function:

$$K(x_i, x_j) = \frac{1}{\left[1 + \left(\frac{2\sqrt{\|x_i - x_j\|^2} \sqrt{2^{\frac{1}{b}}-1}}{\sigma}\right)^2\right]^w} \quad (11)$$

3.2.3. M5P-tree model (M5P)

The M5P, originally introduced for regression problems, represents a genetic algorithm innovation [70]. This method diverges from the traditional approach by focusing on continuous class issues, not the discrete segments, making it particularly adept at handling functions with exceedingly high dimensionality [70], as depicted in Fig. 9a-b. The M5P model effectively harnesses the available data pertaining to each variable within its structure to deduce nonlinear correlations among datasets. At every node of the M5P model, which marks a division in the model tree, a comprehensive evaluation takes place. This encompasses the computation of error estimations and the derivation of crucial parameter insights. In this process, rigorous checks are conducted, including an examination of the default values of inputs and node's classes to identify potential issues. Attributes that have the potential to maximize predicted errors are systematically assessed to gauge their significance within each node's features. It is worth highlighting that the M5P model's underpinnings are firmly rooted in these nodal error estimations. Once the division parameters governing the M5P model's structure have been ascertained, the subsequent construction of the model tree relies heavily on these error predictions [69]. The evaluation of M5P faults relies on variance of the standard class value at nodes [18]. This process typically results in the creation of a complex tree structure due to the successive divisions, which can lead to over complication.

Consequently, the second stage of the process involves pruning the excessively large trees and replacing the pruned sub-trees with linear regression functions [71].

4. Results and discussion

4.1. Correlation analysis

The correlation between variables was analyzed to determine their relationship and inform the design of the predictive model for improved prediction outcomes. To this end, a Pearson correlation matrix was built using Python version 3.7.15 to assess the correlation between independent inputs, as shown in Fig. 10. The presence of multicollinearity among input variables can potentially impact the model and its outcomes, introducing bias. When the predictor variables have a significant correlation ($|r| > 0.5$), multicollinearity is often identified. In this analysis, several pairs of variables exhibit such correlations, as indicated by the heat map. Notably, $\frac{w}{s}$ and $\frac{w}{b}$ (with $r = 0.72$), $\frac{FA}{CA}$ and $\frac{a}{c}$ (with $r = 0.84$), $\frac{FA}{CA}$ and $\frac{w}{c}$ (with $r = 0.82$), $\frac{a}{c}$ and $\frac{w}{c}$ (with $r = 0.74$), $\frac{w}{b}$ and $\frac{w}{c}$ (with $r = 0.64$), as well as CS and STS (with $r = 0.54$) all exceed the 0.5 threshold. This signifies a notably strong positive relationship between these variables. Although no substantial negative correlations were observed among the inputs, the presence of several input pairs with $|r|$ greater than 0.5 indicates the potential for multicollinearity.

4.2. Multicollinearity

Table 4 displays the computed VIF and tolerance values. For all inputs, the tolerance levels exceed 0.1 and the VIF values are less than 10. This observation indicates that there is no significant collinearity among the input variables, assuring the integrity of the prediction models. Among these variables, w/s exhibits the highest VIF value at 8.63 and the lowest tolerance value at 0.12. Notably, w/c and w/b also display elevated VIF values and reduced tolerance values. This suggests that w/c and w/b are likely the most important factors in sensitivity analysis, which is consistent with previous research [69].

4.3. ML analysis

4.3.1. Assessment of ANN-based model

The neural network underwent training and testing phases to predict CS and STS. Constructing the ANN model involved iterative trial-and-error experiments, particularly in determining critical factors such as

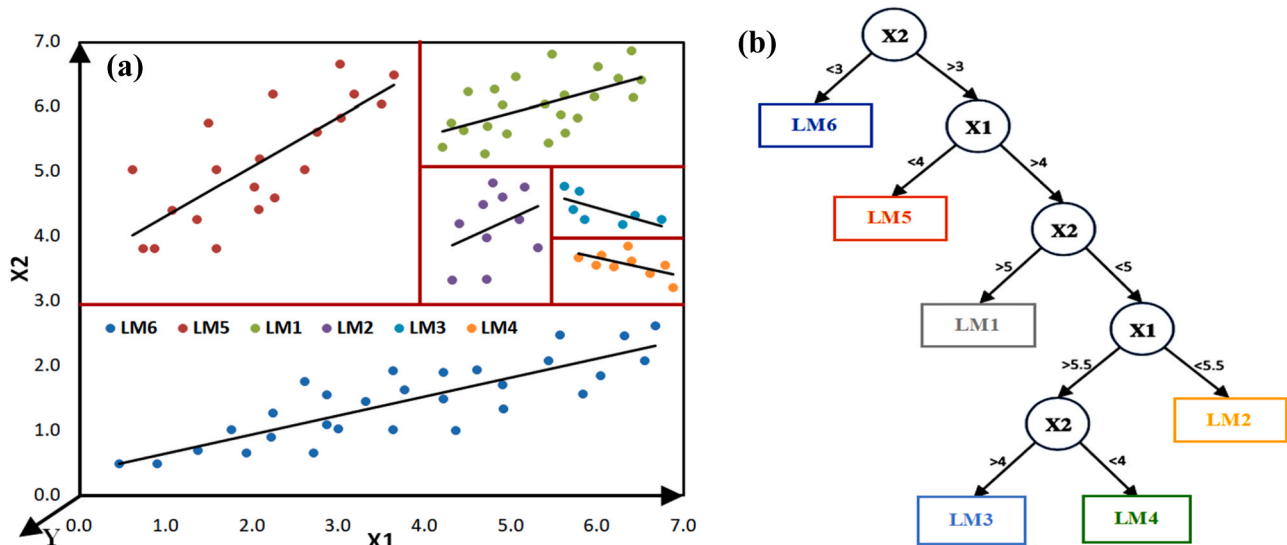


Fig. 9. Schematic of M5PModel: (a) splitting of a 2-D input space and (b) constructing tree into 6 linear models as an example.

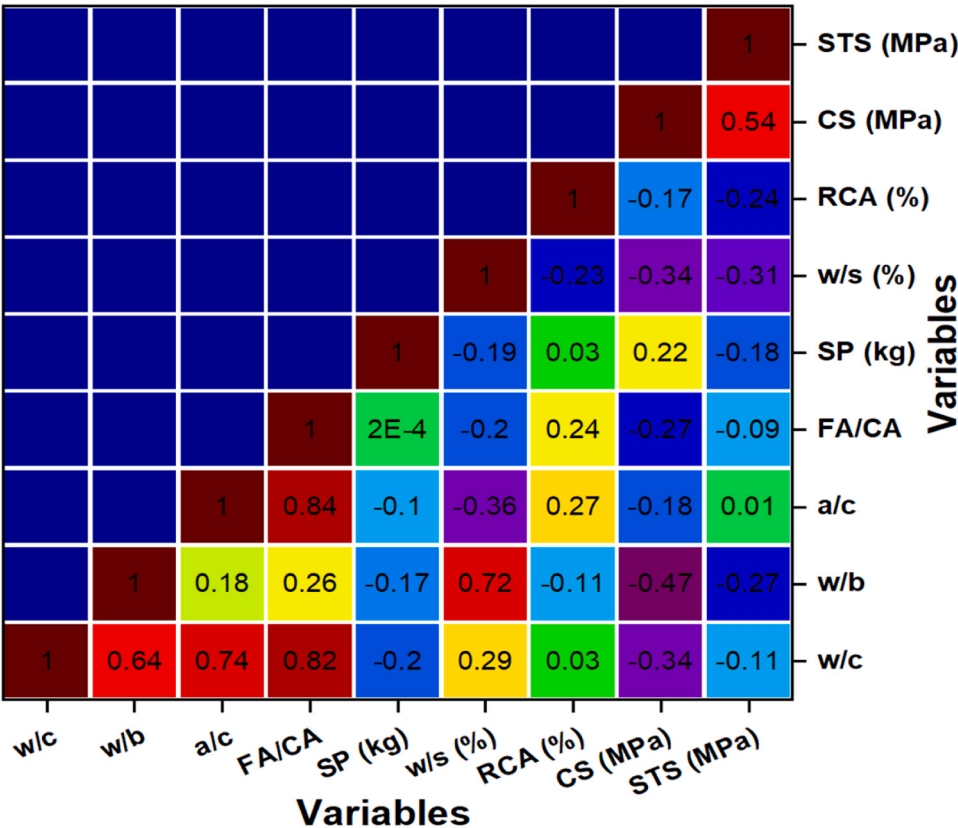


Fig. 10. Correlation coefficient between the variables.

Table 4
VIF and the tolerance values.

		Inputs							Outputs	
		w/c	w/b	a/c	FA/CA	SP(kg)	w/s(%)	RCA(%)	CS(MPa)	STS(MPa)
Collinearity statistics	VIF	4.82	3.91	4.89	4.84	1.40	8.63	1.23	-	-
	1/VIF	0.21	0.26	0.20	0.21	0.72	0.12	0.81	-	-

1/ VIF: the tolerance

the number of hidden layers and neurons within those layers. The finalized ANN model comprises two hidden layers, with the first layer housing five neurons and the second containing two. The experiment used a momentum value of 0.1 and a learning rate of 0.2. Table 5 provides a summary of the ANN model's performance, yielding impressive results. The model attained a high correlation coefficient (CC) of 0.901 and 0.893 for the Training and Testing phases, respectively, in the CS

domain. Additionally, the MAE values were 4.060 and 4.211, and RMSE values were 5.310 and 5.560, respectively. Similarly, for STS, the model exhibited a CC of 0.9275 and 0.913, with MAE values of 0.275 and 0.292, and RMSE values of 0.370 and 0.392 for both stages. Table 5 demonstrates minimal discrepancies between predicted and actual values, with CC consistently exceeding 0.9. These results support the 7–5–2–1 ANN-based model's effectiveness in predicting CS and STS in

Table 5
Assessment of models performance.

CS								
Models	Training stage			Testing stage			COM	Ranking
	CC	MAE	RMSE	CC	MAE	RMSE		
ANN	0.901	4.060	5.310	0.893	4.211	5.560	0.204	1
SVM_PUK	0.942	1.919	4.388	0.821	5.639	8.016	0.274	2
M5P-pruned	0.902	4.402	5.943	0.736	7.214	9.218	0.405	4
M5P-unpruned	0.894	5.055	6.258	0.812	6.834	8.312	0.402	3
STS								
ANN	0.927	0.275	0.370	0.913	0.292	0.392	0.0010	1
SVM_PUK	0.906	0.123	0.404	0.780	0.389	0.658	0.0015	2
M5P-pruned	0.709	0.431	0.757	0.672	0.492	0.773	0.0025	4
M5P-unpruned	0.854	0.318	0.491	0.736	0.402	0.732	0.0019	3

SCC datasets. Figs. S1 and S2 shows the ANN-based model's distribution of actual/predicted CS and STS values for tested concrete.

4.3.2. Assessment of SVM-based model

The employed SVM technique is rooted in statistical theory and leverages a kernel function to tackle intricate and nonlinear regression estimation problems. This supervised approach is versatile, capable of handling both classification and regression tasks. SVM possesses the capability to discern an optimal hyper plane for effectively partitioning data into well-defined categories or making precise predictions in regression contexts. The performance results, as indicated by the higher values in Table 5, underscore the effectiveness of SVM-PUK for prediction. The SVM-PUK-based model exhibits strong performance, yielding a CC of 0.942 and 0.821 for CS, along with MAE values of 1.919 and 5.639 and RMSE values of 4.388 and 8.016 for both training and testing stages. Similarly, for training and testing stages of STS, the model achieves a CC of 0.906 and 0.780, with MAE values of 0.123 and 0.389 and RMSE values of 0.404 and 0.658 for both. These results show that SVM-PUK can accurately predict CS for SCC in both the training and testing datasets. The distribution plots in Fig. S3 and Fig. S4 further illustrate the agreement between predicted and actual strengths.

4.3.3. Assessment of M5P-based model

To identify the most suitable model structure for the dataset, two variations of M5P models, namely pruned and unpruned, were crafted and rigorously evaluated. The outcomes are thoughtfully presented in Table 5. Following the construction of these models, a meticulous assessment was carried out using a set of performance metrics. The notably high CC values featured in Table 6 provide strong evidence of the M5P concrete model's proficiency. For the M5P-pruned model, Fig. S5 and Fig. S6 offer graphical representations of the distribution and agreement between predicted and actual strengths (CS and STS) of SCC during both the training and testing phases. Notably, the M5P-pruned model excelled in forecasting CS, achieving CC values of 0.902 and 0.736, alongside MAE values of 3.402 and 7.214 and RMSE values of 5.943 and 9.218 for the training and testing phases, respectively. However, for STS predictions, the M5P-pruned model exhibited subpar performance across the board, with lower CC (0.709), higher RMSE (0.757), and elevated MAE (0.431) values observed during both the training and testing phases. On the other hand, the M5P-unpruned model fared well in predicting CS, attaining MAE values of 5.055 and 6.834, CC values of 0.894 and 0.812, and RMSE values of 6.258 and 8.312 for the training and testing stages, respectively. Similarly, the M5P-unpruned model exhibited robust performance for STS, with MAE values of 0.318 and 0.402, CC values of 0.854 and 0.736, and RMSE values of 0.491 and 0.732 for the training and testing phases, respectively. Fig. S7-S8 provides visual representations of the distribution and correlation between predicted and actual strengths of SCC for the training and testing phases of the M5P-unpruned model. In summary, it is evident that the M5P-unpruned model outperformed the M5P-pruned model in forecasting both the CS and STS of SCC. These assessment

findings collectively affirm the superior prediction capabilities of the M5P-unpruned model for CS and STS compared to its pruned counterpart.

4.3.4. Comparison of results

In this work, several ML techniques (ANN, SVM, M5P-pruned, and M5P-unpruned) were utilized to predict the CS and STS of SCC. The following input variables were utilized to predict STS and CS: RCA%, w/c, w/b, a/c, FA/CA, w/s, and SP. Table 6 displays the performance assessment metrics derived from the models employed throughout the training and testing stages. Choosing the most accurate model for forecasting the CS of the SCC, as shown in Table 5, posed a significant difficulty. The SVM-PUK-based model performs better than others for CS prediction, with a training phase CC of 0.942, MAE of 1.919, and RMSE of 4.388. With a CC of 0.893, MAE of 4.211, and RMSE of 5.56 for the testing phase, the ANN-based model surpasses the competition. On the other side, for STS prediction, the evaluation of the models revealed that the ANN-based models outperformed the SVM, M5P-pruned, and M5P-unpruned models. With CC values of 0.927 and 0.913, MAE values of 0.275 and 0.292, and RMSE values of 0.370 and 0.392 for the training and testing phases, respectively, the ANN-based model surpasses competitors. A COM was used to rank the models, and the results showed that for the provided datasets, the ANN model outperformed the SVM with the polynomial kernel (SVM-PUK), M5P-pruned, and M5P-unpruned models, in that order.

The errors were analyzed and documented in Fig. 11a-b to assess the accuracy of the used models in predicting the CS and STS of SCC. As shown in Fig. 11a-b, the error bandwidth is less for the ANN and SVM-PUK models when compared to other models proposing interpretations for CS and STS. There is an actual connection between the observed and predicted variables since the values on these graphs are concentrated along the line of complete agreement. With a larger CC and less insignificant CS errors, the ANN-based model performed comparably to earlier linear regression models across all phases. Both the CS and STS models' efficacy is displayed in Figs. 12a-b and 13a-b, respectively. These figures demonstrate that the ANN model's predicted variables are in close proximity to the actual dataset, resulting in a small error bandwidth ($\pm 20\%$ error line for CS and STS).

Additionally, the SI value was calculated to evaluate the effectiveness of the predictive models. Fig. 14a-b demonstrates that the ANN-based models accurately predicted the strengths (CS and STS) with SI values close to the 10% line. On the other hand, the SVM-based model achieved proper predictions for the CS and STS of SCC in both stages, although with slightly higher SI values. Specifically, the SI values for CS were 0.0999 and 0.1835 in the training and testing stages, respectively. Similarly, the SI values for STS were 0.1184 and 0.1931 in the training and testing stages, respectively. These values were all below 20%, indicating the adequacy of the SVM-based model. Furthermore, Fig. 14a-b reveals that the M5P-unpruned-based model adequately predicted CS in training/testing stages, but its performance in predicting STS was somewhat satisfactory. Therefore, based on the comparison and evaluation of the ML techniques, the ANN-based models performed the best in predicting the CS and STS of SCC. The SVM-based model showed promising results, although with slightly higher SI values. However, despite its satisfactory performance in predicting CS, the M5P-unpruned-based model demonstrated limitations in predicting STS.

While the comparison of actual and predicted values for CS across various modeling approaches revealed insignificant variations, a closer examination of Table 6 indicates that the p-value associated with the ANN is notably close to 1. This close proximity to 1 indicates that the null hypothesis, which is used to predict the CS of concrete made with RCA, is valid. Despite the lack of significant differences, this finding highlights the specific strength of the ANN-based model in capturing and predicting the CS of concrete with RCA. These findings are similar to what Arora et al. observed in 2019 [72]. Furthermore, when evaluating the performance in predicting STS, the ANN-based model outperforms

Table 6

Results of static CS and STS via single factor ANOVA test for ANN-based, SVM-based, M5P-pruned-based, and M5P-unpruned-based models.

CS			
Models	F	P-value	F crit
ANN-based model	0.00839780	0.9270284	3.8638014
SVM-based model	0.06822321	0.8819059	3.8638014
M5P-pruned-based model	0.08052821	0.7767234	3.8638014
M5P-unpruned-based model	0.07100452	0.8347311	3.8638014
STS			
ANN-based model	0.00543237	0.9491983	3.8638014
SVM-based model	0.02591125	0.8721486	3.8638014
M5P-pruned-based model	0.05374219	0.6013755	3.8638014
M5P-unpruned-based model	0.03421617	0.7883061	3.8638014

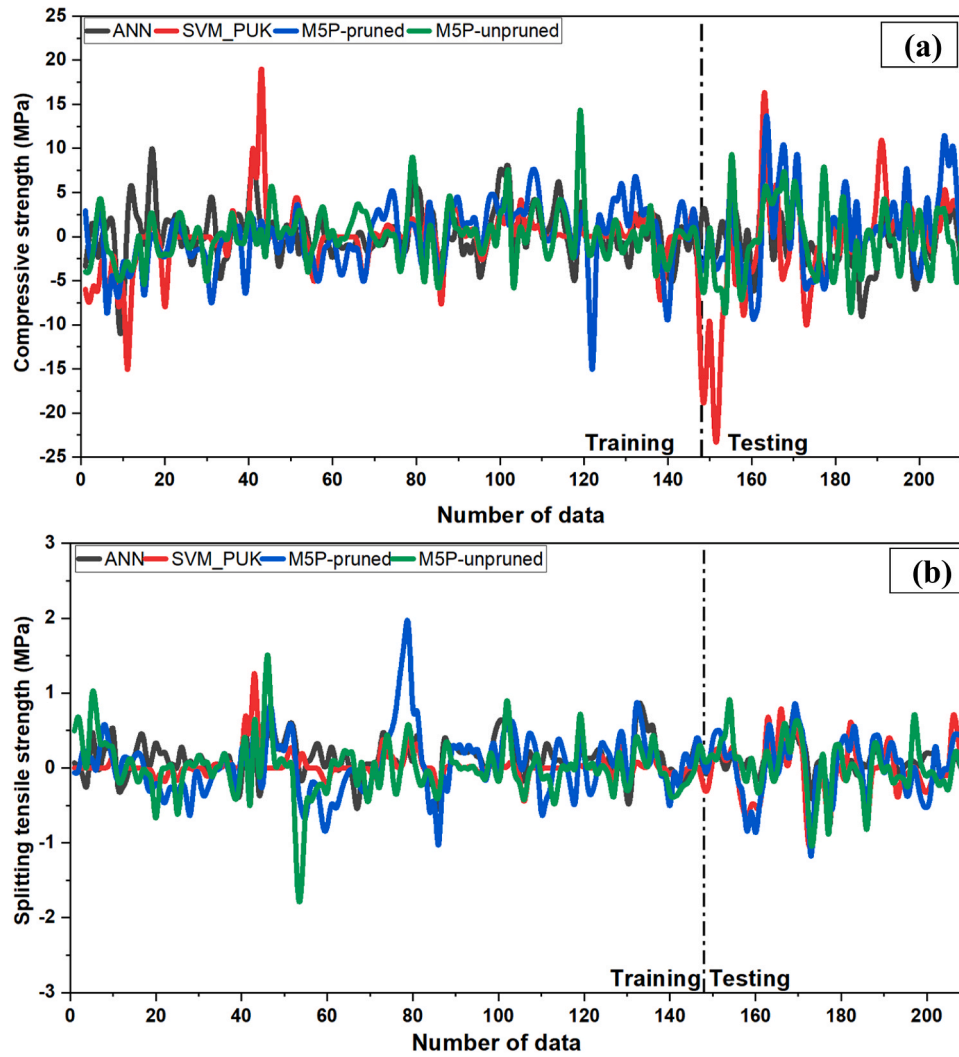


Fig. 11. Relative errors of models for: (a) CS and (b) STS in training and testing datasets.

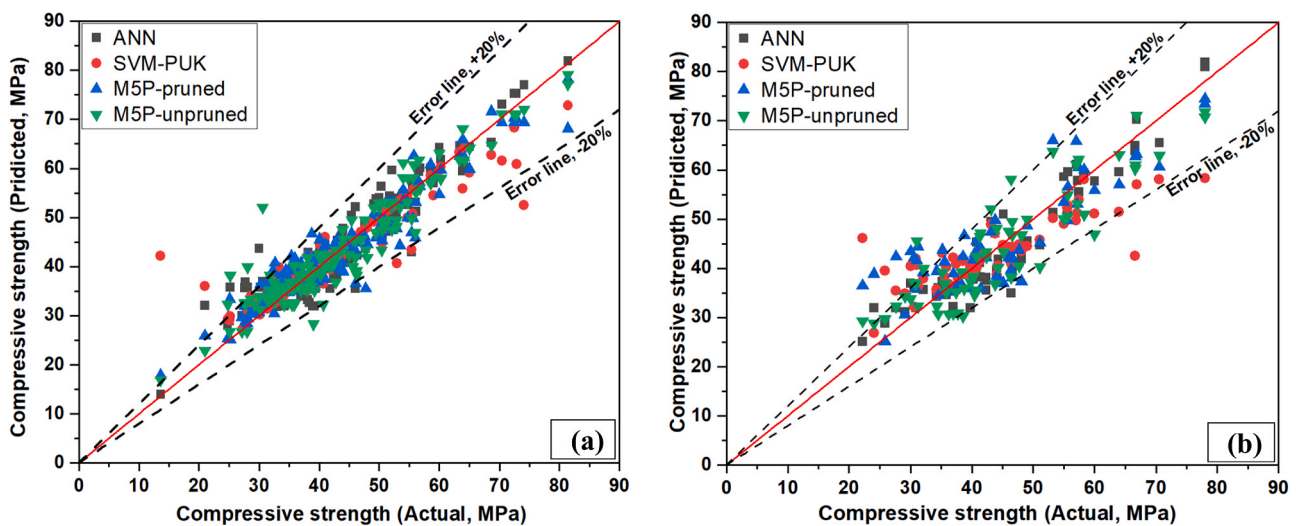


Fig. 12. Compression strength comparison among utilized approaches during, (a) Training, and (b) Testing phases.

the SVM and M5P-based models, as evidenced in Table 6. In this case, the p-values are also explicitly mentioned as higher than the one in CS, but the emphasis on the superior predictive capabilities of ANN implies

that the null hypothesis remains valid. The insignificant difference observed between actual and predicted values for splitting tensile strength, as determined by single-factor ANOVA (referenced in Table 6),

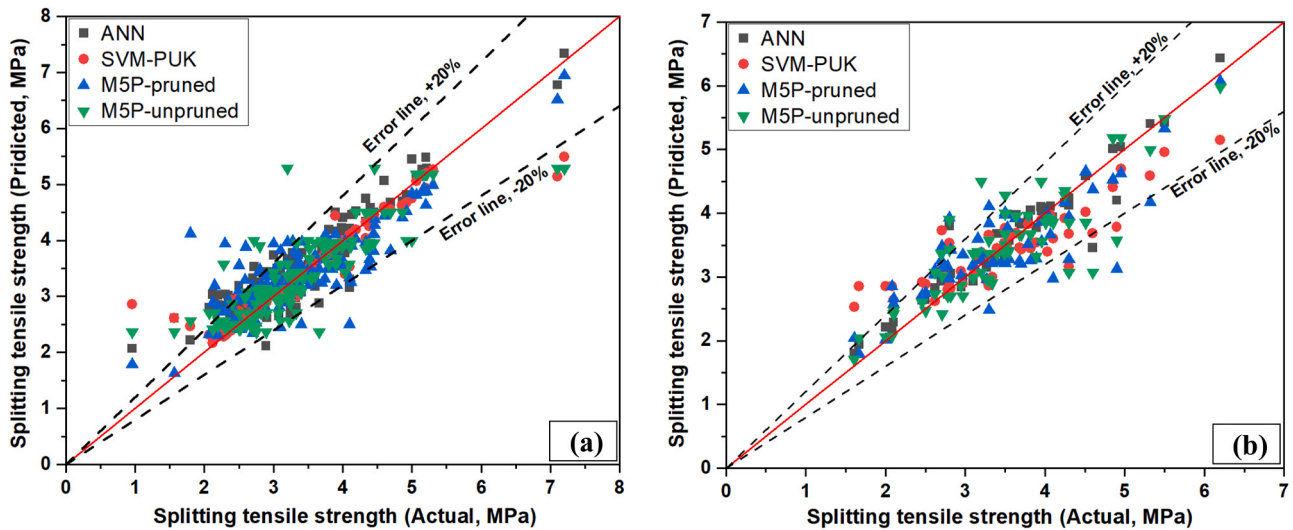


Fig. 13. Splitting tensile strength comparison among utilized approaches during, (a) Training, and (b) Testing phases.

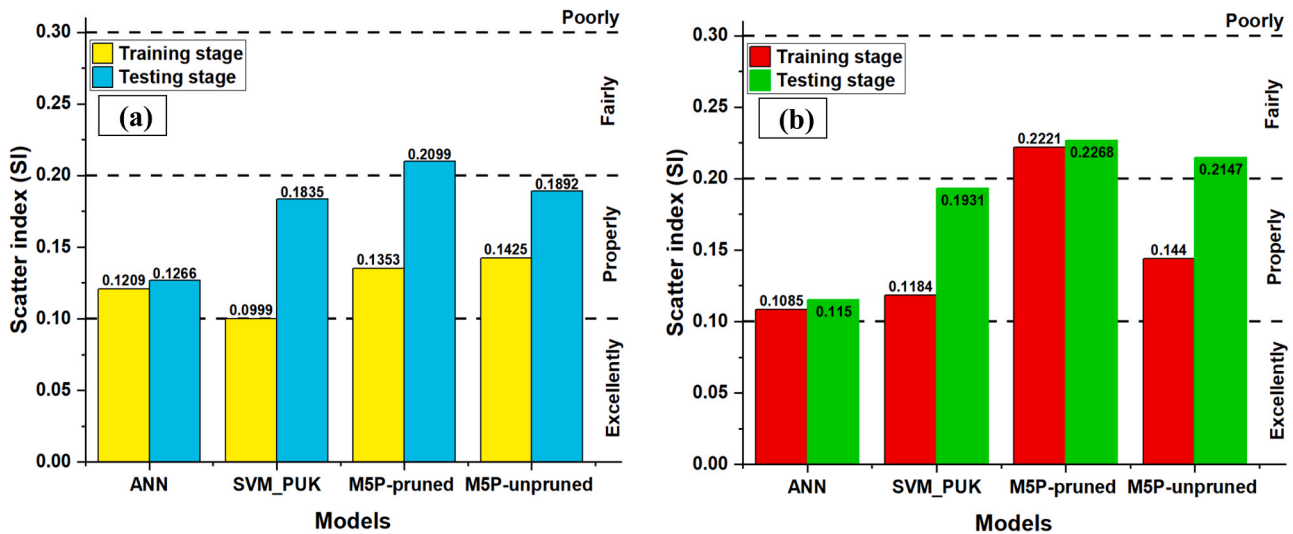


Fig. 14. Scatter index plots illustrating model comparisons for both training and testing stages: (a) CS models and (b) STS models.

underscores the comparable accuracy of ANN, SVM, and M5P-based models for these new presented parameters. Despite the lack of statistical significance, the results suggest that the ANN-based model stands out in its predictive accuracy for STS in comparison to the other modeling techniques employed.

4.3.5. Comparing results with contemporary works

Table 7 displays the results of a statistical comparison of the models developed in this study with those found in the literature, using the CC, MAE, and RMSE on both the actual and predicted datasets. The findings reveal that, in comparison to models from prior research, all the models presented in this study exhibit higher CC values and lower MAE and RMSE values. A distinct pattern becomes evident after carefully comparing the model values found in this investigation with those documented in the literature. The models created in this work routinely perform better than those from earlier studies. The higher CC values, which show a stronger correlation between the predicted and actual values, provide evidence for this. Additionally, the lower MAE and RMSE values signify reduced errors in the predictions made by the models developed in this study. Notably, the M5P linear model, specifically crafted in this study, stands out as particularly robust. It surpasses

the performance of models found in the literature, showcasing superior predictive capabilities. This is exemplified by CC values of 0.86 for CS and 0.80 for STS.

4.3.6. Monte-Carlo simulation (MCS) of uncertainty

MCS method is employed to model and address uncertainty and randomness within a given model. Interestingly, it was the military that pioneered the adoption of MCS, primarily employing it to tackle probabilistic events [79]. Predicting the CS and STS presents formidable challenges and uncertainties, encompassing factors such as uncertainties associated with input predictors and variations in model parameters. To grapple with these complexities, MCS is seamlessly integrated into predictive models, namely ANN, SVM, and M5P, which leverage CS and STS values.

The results of this study, portrayed in Fig. 15a-b, encompass critical metrics including the Mean Absolute Deviation (MAD), the median of predicted CS and STS, as well as the range of uncertainty. Remarkably, the positive values in the mean estimation error signify that the predicted CS and STS derived from these predictive models tend to exceed experimental values. Moreover, the study illuminates that the ANN and M5P-unpruned models yield the smallest (15.038%) and largest

Table 7

Comparison of current study's models with literature.

References	Predict models	Input no.	Input type	MAE	RMSE	CC
Present study	ANN	7	Non-dimensional	4.14	5.44	0.90
Present study	SVM	7	Non-dimensional	2.35	5.49	0.90
Present study	LM of M5P:	7	Non-dimensional	5.95	7.29	0.86
[73]	$CS = 19.8447 * \frac{w}{c} - 98.0111 * \frac{w}{b} - 3.7208 * \frac{A}{c} + 4.0341 * \frac{FA}{CA} + 0.5406 * SP + 1.2663 * \frac{w}{s} (\%) - 0.0624 * RCA(\%) + 78.9773$	8	Dimensional	7.61	-	0.65
	$CS = -0.0902 * W + 0.1065 * F + 0.1279 * C + 0.1089 * A + 0.3079 * S + 0.0229 * CA + 0.03116 * FA - 50.77$	14	Dimensional and non-dimensional	11.02	-	0.87
[74]	ANN	8	Dimensional	7.43	7.73	0.78
[75]	SVM	7	Dimensional	0.84	1.20	0.89
[76]	M5P	7	Non-dimensional	0.28	0.38	0.92
STS	ANN	7	Non-dimensional	0.26	0.53	0.85
Present study	SVM	7	Non-dimensional	0.36	0.61	0.80
Present study	LM of M5P:	7	Non-dimensional	0.36	0.61	0.80
[77]	$CS = 0.1627 * \frac{w}{c} + 0.1196 * \frac{w}{b} + 0.0072 * \frac{A}{c} - 0.0233 * \frac{FA}{CA} + 0.0033 * SP - 0.0556 * \frac{w}{s} - 0.0018 * RCA(\%) + 3.6652$	4	Dimensional and non-dimensional	-	0.84	0.64
	Global MNR equation	14	Dimensional and non-dimensional	-	-	0.82
[78]	$CS = -0.009 * R - 7.259 * \frac{w_{eff}}{c} - 1.288 * \frac{D_{max,r}}{D_{max,n}} - 0.322 * \frac{a}{c} + 0.005 * R * \frac{w_{eff}}{c} + 0.003 * R * \frac{D_{max,r}}{D_{max,n}} - 0.001 * R * \frac{a}{c} + 2.107 * \frac{w_{eff}}{c} * \frac{D_{max,r}}{D_{max,n}} + 0.466 * \frac{w_{eff}}{c} * \frac{a}{c} - 0.021 * \frac{D_{max,r}}{D_{max,n}} * \frac{a}{c} + 7.598$	14	Dimensional and non-dimensional	-	-	0.82
[78]	ANN	7	Dimensional	0.34	0.43	0.82
[78]	Tree-based	7	Dimensional	0.34	0.43	0.82

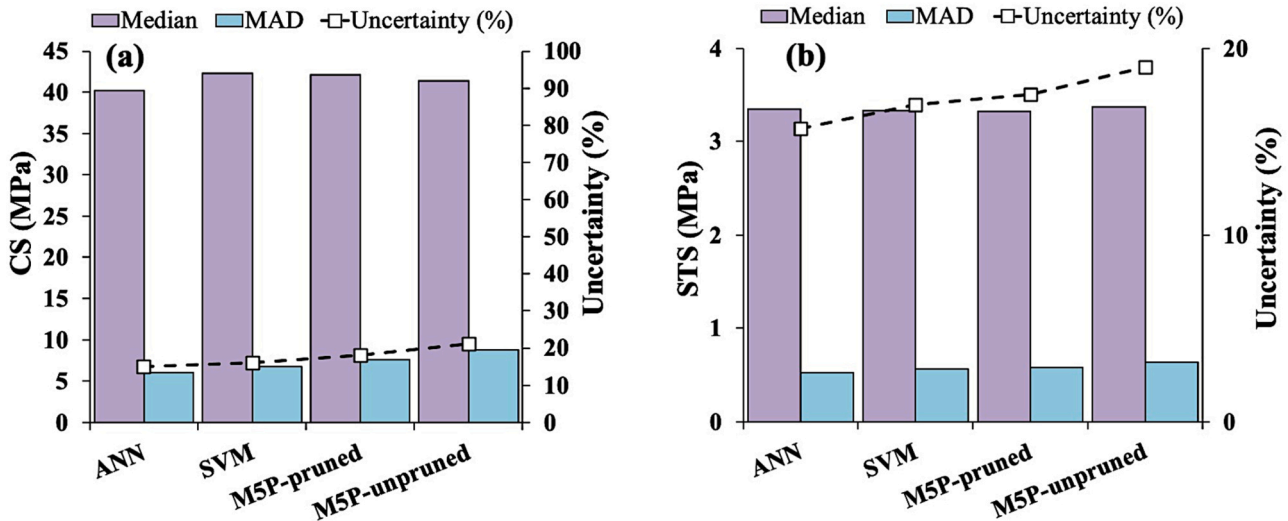


Fig. 15. Uncertainty analysis of models: (a) CS and (b) STS.

(21.154%) uncertainty bands for CS, respectively. Similarly, for STS, the ANN and M5P-unpruned models produce the most slender (15.701%) and expansive (19.008%) uncertainty bands, respectively. It is worth noting that all of these percentages fall within an acceptable range (threshold of 35%).

4.3.7. Sensitivity analysis

Using the ANN model of testing datasets, a sensitivity test was performed to determine the relevant input factors impacting CS and STS by deleting one input parameter at a time. The effect of each input variable on CS and STS was evaluated in terms of its influence on the outputs via

CC, MAE, and RMSE, as shown in Fig. 16 (CS) and Fig. 17 (STS). The results indicate that, based on this dataset, w/c has a significant role in predicting the CS and STS using the ANN model of 7–5–2–1, followed by w/b and w/s, which endorses the findings of Alyaseen et al. [79]. Meanwhile, the remaining design parameters were ranked as shown in Fig. 16 (CS) and 17 (STS).

4.3.8. Parametric evaluation of SCC with RCA on CS and STS

A parametric analysis was conducted to assess the presented ANN model's performance, focusing on experimentally influential factors such as FA/CA, RCA, a/c, w/s, w/c, SP, and w/b. This evaluation aimed

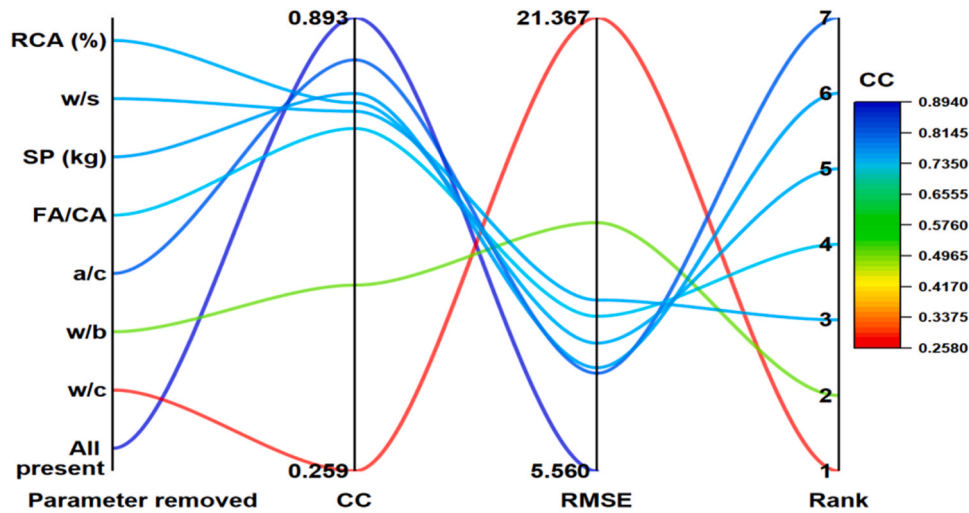


Fig. 16. Sensitivity analysis for CS using ANN.

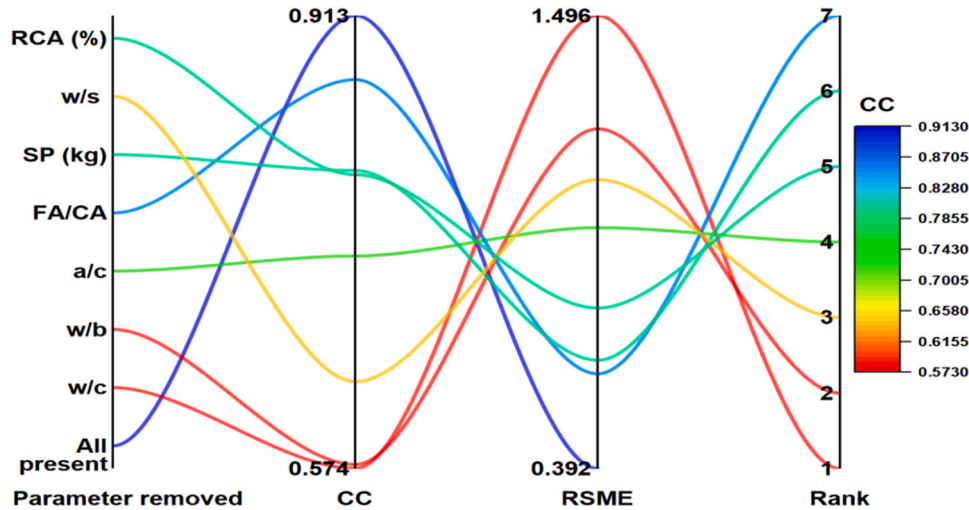


Fig. 17. Sensitivity analysis for STS using ANN.

to determine the ANN-based model's quality and validity when applied to RCA SCC mixtures. In this investigation, the most effecting input variables on CS and STS were considered as variables (Section 4.3.7) while keeping other experimental factors constant based on their mean dataset values [79]. This approach was undertaken to isolate the effects of specific factors on CS and STS, contributing to the observed linearity. It's worth noting that the mechanical properties of mixtures are heavily influenced by both mix design and material composition, with even minor material composition variations exerting significant impacts. The works conducted with the optimal content produced a more uniform mixture and, as a result, enhanced CS and STS. Overall, the model accurately predicts all trends with smooth curves. Specifically, Fig. 18a-d and Fig. 19a-d depict the expected decrease in CS and STS values with the increase of the w/c and a/c ratios, which align with the finding of many researchers [80–83]. Moreover, the fact that excessive w/c and a/c ratios in SCC with RCA may cause segregation problems [84]. Perversely, as shown in Fig. 18a-d and Fig. 19a-d, the expected increase in w/b and w/s values leads to an increase in CS and STS [79–84]. To elaborate further, the observed linearity aligns with established theories related to some researchers [80,81]. For example, the expected decrease in CS and STS values with the increase of w/c and a/c ratios is consistent with. Similarly, the anticipated increase in CS and STS with higher w/b and w/s values is in accordance with [provide

theoretical support].

5. Limitations and future orientation

The application of ML approaches is beset by several notable limitations, accompanied by potential avenues for future research enhancement. One significant drawback lies in the inherent opacity of ML algorithms, which often defy straightforward explanations of their decision-making processes. ML algorithms can be likened to enigmatic black boxes, receiving inputs and yielding outputs without elucidating the intricacies of their computations. This opacity raises concerns about interpretability and accountability. Furthermore, ML is susceptible and vulnerable to the fallacy of GIGO ("garbage in, garbage out"). Essentially, the quality of the outcomes produced by ML algorithms is intimately tied to the quality of the input data. Any inaccuracies or errors in the labelling of the input data will invariably manifest in the algorithm's outputs, compromising the reliability of the results.

To address these limitations and advance the field, future studies employing ML approaches for predictive tasks can benefit from several strategies. First and foremost, researchers should explore the optimization of hyper parameters for their chosen models. Fine-tuning these parameters can significantly enhance model performance and generalization. Additionally, researchers can employ convergence curves based

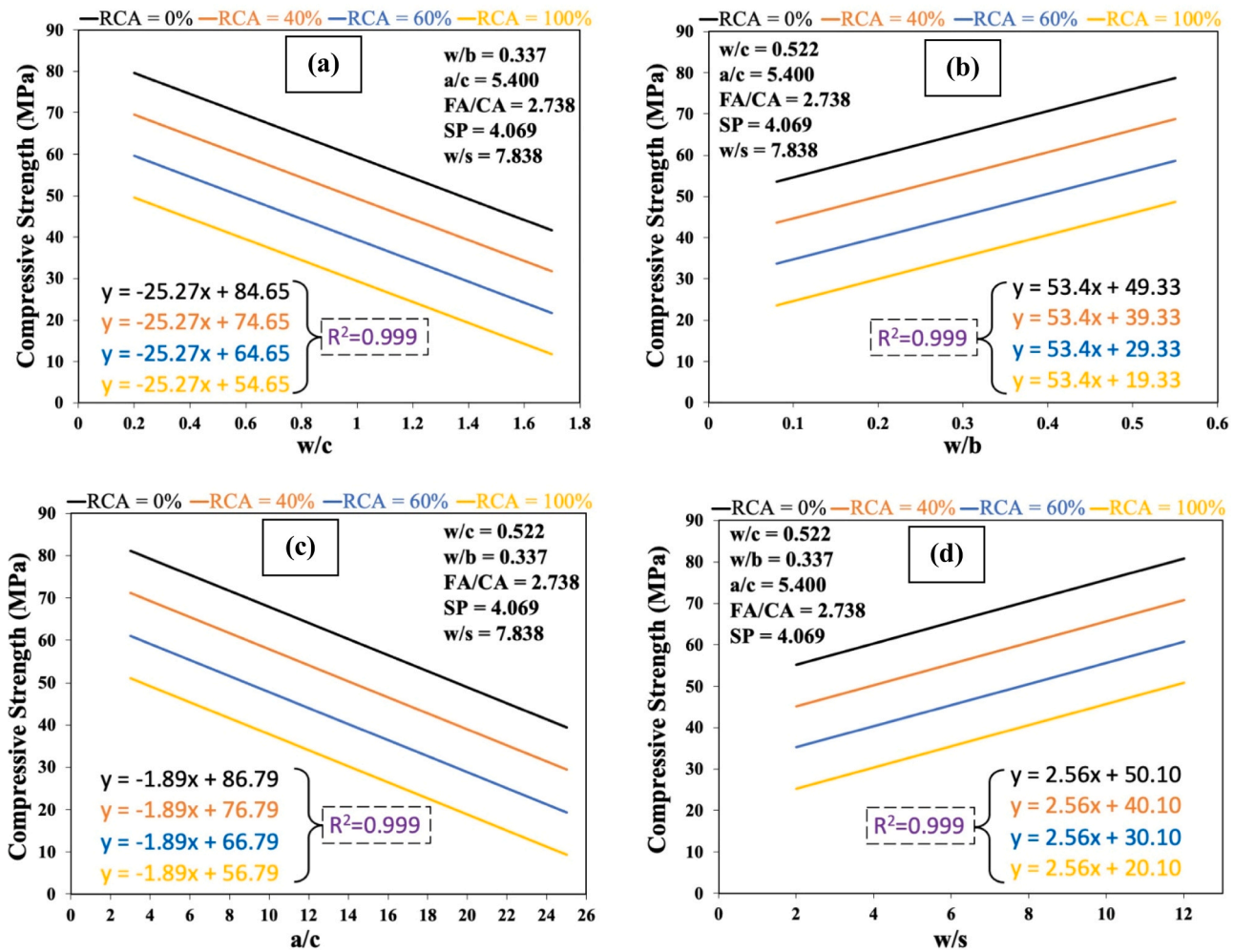


Fig. 18. Parametric study of SCC mixtures at different RCA% using ANN model: (a) CS vs w/c, (b) CS vs w/b, (c) CS vs a/c, and (d) CS vs w/s.

on RMSE to assess the model's convergence behavior. This enables a more thorough understanding of how the model learns and improves over time. Moreover, new evaluation metrics, such as prediction interval and A-10index, can be incorporated to provide a comprehensive assessment of model accuracy and uncertainty.

In summary, addressing the challenges posed by ML's opacity and sensitivity to data quality requires a multifaceted approach involving both improved model optimization techniques and more sophisticated evaluation methodologies. These efforts can pave the way for more reliable and interpretable ML applications in various domains.

In a related context, regarding the potential influence of processing techniques (e.g., Mix Design, Mixing Techniques, Placing Techniques, Compaction Techniques, Curing Techniques, Finishing Techniques or Surface Treatment, and Admixtures) on the properties of concrete. It is indeed an important factor that can impact the end-use application properties. In this study, the authors focused primarily on comparing the predictive abilities of three different machine learning techniques in evaluating the CS and STS of a 28-day-old SCC incorporating RCA. While the significance of processing techniques should be acknowledged, the current investigation specifically targeted the machine learning aspect of predicting the concrete properties of admixtures. However, the value of considering processing techniques in future research endeavours must be recognized to provide a more comprehensive understanding of factors influencing concrete properties.

6. Conclusions

This study proposes a methodical data analytics framework tailored to the examination of experimental datasets containing SCC samples characterized by varying RCA proportions. While other studies primarily use computational approaches to investigate the behavior of SCC strengths (CS and STS) based on dimensional mix design variables, this study aims to thoroughly analyze the observed variables related to the tested datasets of non-dimensional elements (design parameters) from the literature and evaluate their strength based on analysis indicators. A systematic methodological framework was proposed, and multiple methods, including PCC, VIF (statistical), ANN, SVM, M5P (ML) and sensitivity analysis, were used to get comprehensive and profound insights from the dataset. This approach is significantly distinct from other experimental-based investigations and allows for a thorough evaluation of the tested datasets and the strength of the observed variables. The results and insights of the analysis are presented below:

- The dataset for this study consisted of 210 SCC readings with various RCA percentages and included the CS and STS. The dataset was compiled from as many tests of SCC as were attainable in order to provide a comprehensive analysis. Among the CS and STS, the overlapping and compatible datasets for the seven inputs were tested at 28 days for the CS and STS, which have been pinpointed and selected from the literature to support analysis.
- This study initiated with a comprehensive examination of the datasets, commencing with a meticulous descriptive analysis. A pivotal

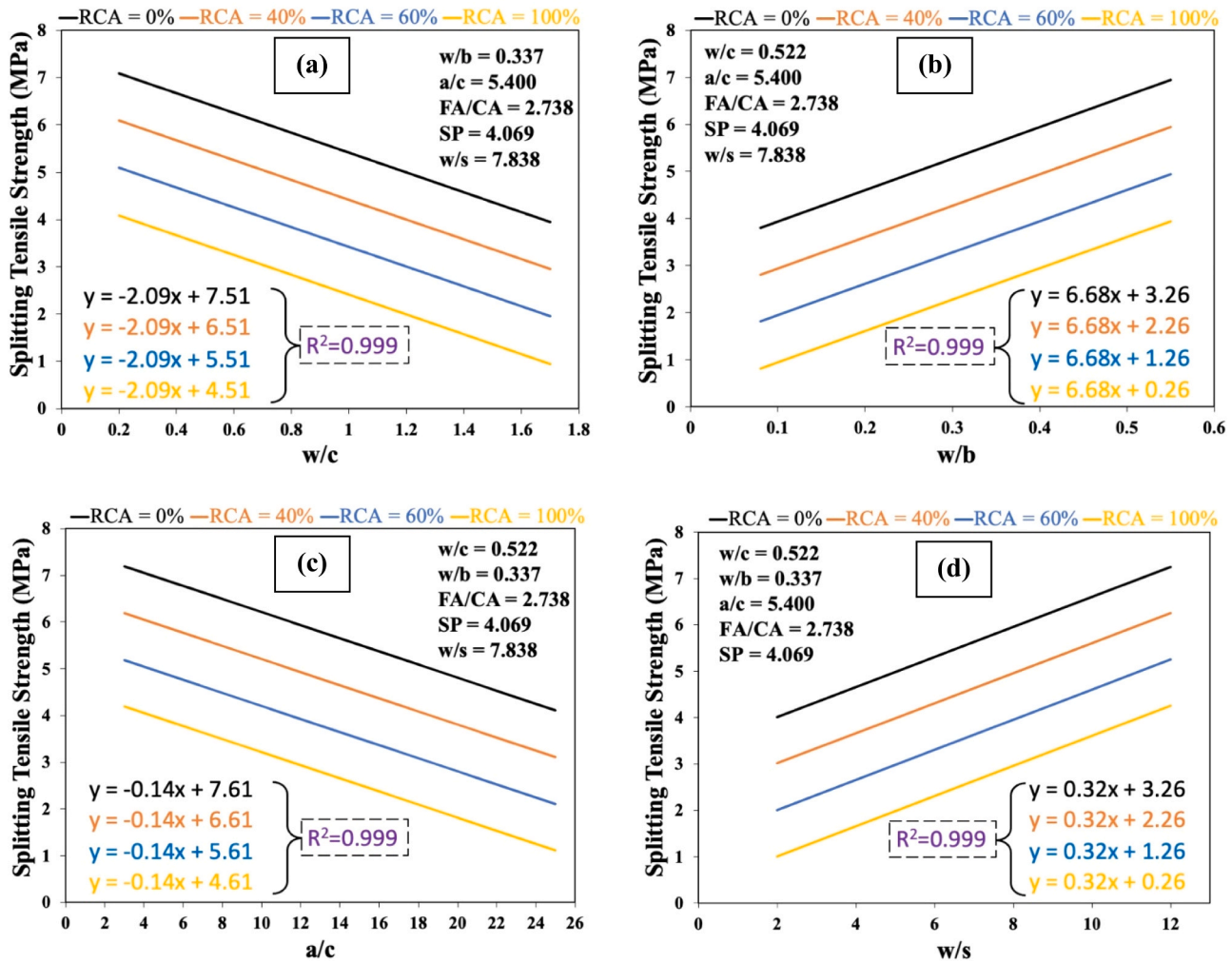


Fig. 19. Parametric study of SCC mixtures at different RCA%: (a) STS vs w/c, (b) STS vs w/b, (c) STS vs a/c, and (d) STS vs w/s.

facet of this analysis involved conducting a correlation analysis utilizing PCC. The PCC served as a valuable tool for discerning variable groups characterized by robust interrelationships. The outcomes of this analysis were unequivocal: several pairs of variables exhibited correlations exceeding 0.5. Specifically, the variables w/b and w/c, FA/CA and a/c, w/b and w/s, w/c and a/c, FA/CA and w/c, and CS and STS demonstrated pronounced positive correlations. These findings illuminate that alterations in one variable are strongly and consistently associated with corresponding changes in the other variable in the same direction. Conversely, the analysis indicated a conspicuous absence of robust negative correlations among variables. In simpler terms, the relationships between the remaining variables were either non-existent or exhibited only minimal associations.

- The independent input variables exhibited VIF and tolerance values below 10 and above 0.1, respectively. These findings suggest the absence of significant collinearity among the inputs, which could otherwise have potentially distorted the prediction models.
- In this study, ANN, SVM, and M5P models were used to predict the CS and STS of SCC. The results clearly show that, in terms of predicted accuracy, the ANN model performed better than the other models. This superiority is evident through its higher CC, lower MAE, RMSE, and smaller SI values. In contrast, the M5P-pruned-based model, based on the used dataset, exhibited only moderate predictive capabilities, particularly for CS in SCC. Thus, it can be concluded

that the most suitable and effective technique for accurately predicting the CS and STS of SCC is the ANN model.

- Among the various machine learning models employed in this study, the 7-5-2-1 ANN-based model demonstrated superior predictive capabilities for CS and STS in SCC. The high values of CC, coupled with the low MAE and RMSE, establish the credibility of this model. These findings collectively affirm that the 7-5-2-1 ANN-based model is proficient in forecasting CS and STS in SCC, relying on the information derived from the seven non-dimensional elements, which represent essential design parameters extracted from the literature.
- ANN and M5P-unpruned models exhibit remarkable precision, producing the smallest (15.038%) and highest (21.154%) uncertainty margins for CS, respectively. Likewise, for STS, the ANN and M5P-unpruned models unveil the smallest (15.701%) and highest (19.008%) uncertainty intervals. Thus, these percentages all align within the accepted range (35%) for model uncertainty estimation, underscoring the reliability of these predictive approaches.
- Via the ANOVA analysis, it is confirmed that there are insignificant differences between the results of the ML-based models.
- The comparison conducted between the findings of predictive models in this work and contemporary works in the literature underscores the efficacy of the models, positioning them as advancements in predictive accuracy.
- The sensitivity analysis results highlight that the sequence of significance lies with w/c, w/b, and w/s when it comes to predicting CS and STS in SCC using the ANN-based model. It is reasonable to

attribute their prominence in the sensitivity analysis to their high VIF values and lower tolerance levels, which underscore their pivotal role in this analysis.

- The anticipated outcome involves a reduction in CS and STS as the w/c and a/c ratios increase. It is worth noting that overly high w/c and a/c ratios in RCA-infused SCC can potentially trigger segregation issues. Conversely, an expected increase in w/b and w/s ratios is associated with an increase in CS and STS.

Future studies should delve into deep learning investigations employing a variety of hybrid ANN-based techniques. Moreover, there is a pressing need for further research on the failure mechanisms inherent in RCA concrete, particularly in the context of standard and high-strength grades. Furthermore, the potential influence of processing techniques on the properties of concrete should be studied via ML-based approaches by including these processing techniques as input parameters. This research holds significance due to its potential impact on the sensitivity of CS and STS predictions for high-strength grades of RCA concrete.

CRediT authorship contribution statement

Singh Tej: Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Alyaseen Ahmad:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kumar Navsal:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Poddar Arunava:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Lee Daeho:** Writing – review & editing, Formal analysis. **Sihag Parveen:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Kumar Raj:** Writing – review & editing, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.mtcomm.2023.107970](https://doi.org/10.1016/j.mtcomm.2023.107970).

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